# Same Words, Different Bets

Human vote-distribution supervision for sentiment classification under scarce labels

**Research question.** When only a small labeled sample is available, does training against empirical human vote proportions improve alignment with held-out human judgments compared with majority-label cross-entropy (CE)? We compare human soft targets with majority CE and generic label smoothing across DynaSent Round 1, DynaSent Round 2, and SentiMP-En.

## Method

For each text, the soft target is the observed proportion of annotator votes across the project's four sentiment classes. The primary outcome is per-example Jensen–Shannon (JS) divergence in bits against the human vote distribution (lower is better). Label budgets are 25, 50, 100, 200, and 300. The primary comparison is $\Delta JS = JS_{soft} - JS_{majority\ CE}$; negative values favour soft-target supervision. DynaSent uses 10 seeds per budget; the supplied SentiMP-En primary test table uses 5 seeds. Null-augmented and tie-inclusive variants are reported separately because they change training composition.

## Primary findings

Mean JS divergence is lower with human soft targets in 15 of 15 matched dataset/round-by-budget comparisons. However, paired seed-level exact sign-flip tests with Holm correction across all 15 comparisons identify only 1 comparison at p<0.05: DynaSent Round 2 budget 300 (adjusted p=0.029). This supports a consistent directional trend, with limited corrected inferential evidence.

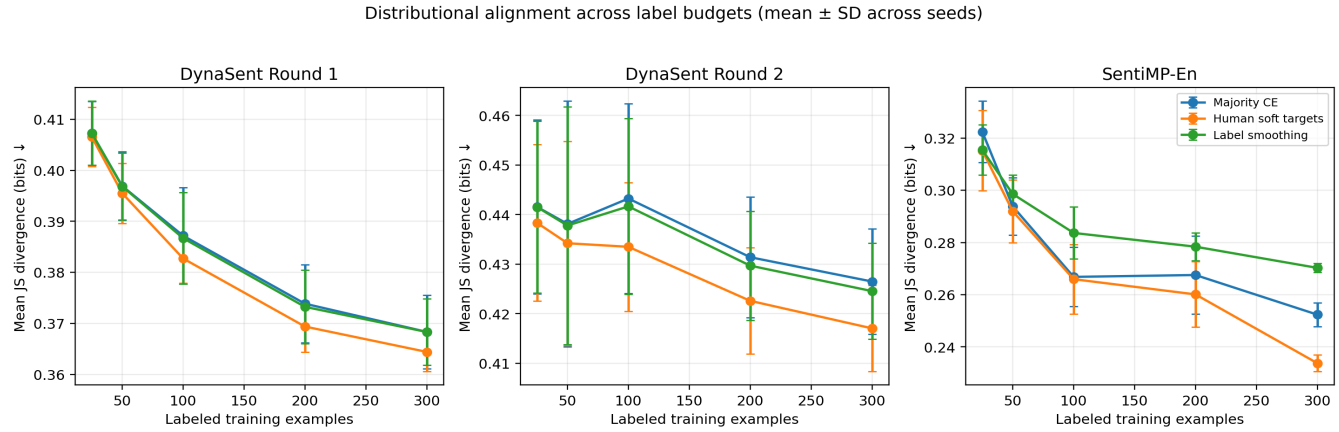



Figure 1. Mean JS divergence by label budget; error bars show standard deviation across seeds, not confidence intervals.

## Primary contrast

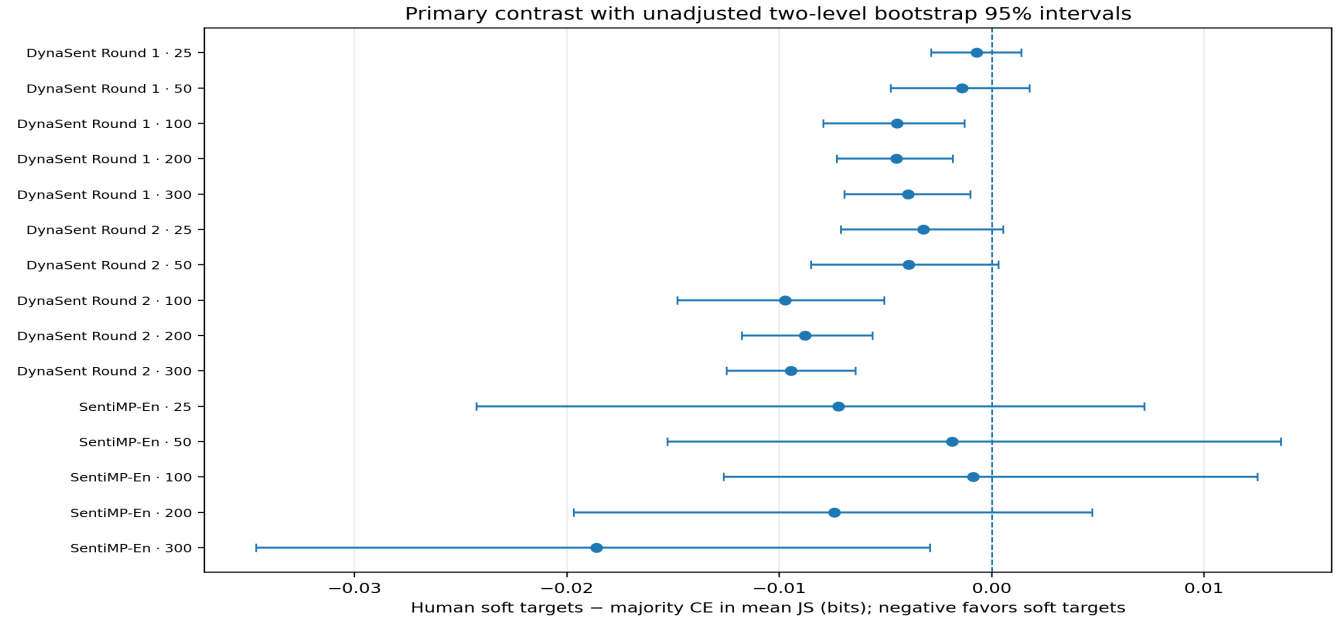



Figure 2. Human-soft-target minus majority-CE JS difference. Points and unadjusted two-level bootstrap intervals are distinct from the seed-level Holm-adjusted tests.

## Interpretation, limitations, and next steps

The results suggest that preserving annotator disagreement can improve distributional alignment under scarce supervision. This is not evidence that soft targets are universally superior: majority-label accuracy and macro-F1 are secondary outcomes and their effects are mixed across settings. Generic label smoothing is an important comparator because it smooths labels without preserving each example's empirical vote distribution.

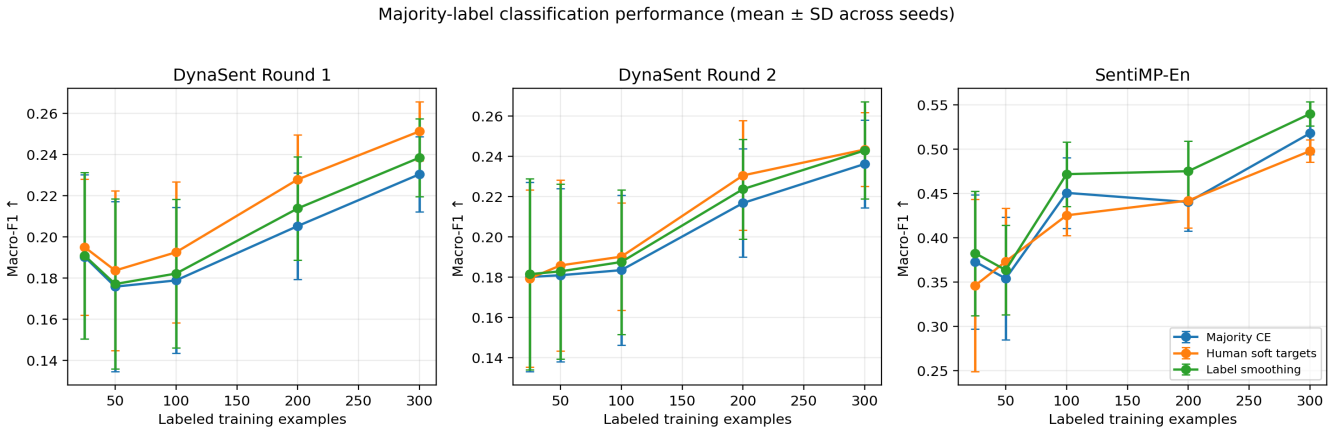



Figure 3. Macro-F1 across budgets. Classification performance should be interpreted separately from distributional alignment.

## Statistical and reporting caveats

• Holm-adjusted p-values are from paired seed-level exact sign-flip tests; bootstrap intervals are unadjusted and quantify a different uncertainty component.

• The analysis is exploratory because the results have already been inspected. A prospective cycle with a frozen protocol is needed for confirmatory claims.

• The null-augmented DynaSent condition and tie-inclusive SentiMP-En condition change training composition and are not pure loss-function comparisons.

• The supplied Tiny-BERT pilot (100 labels, five seeds) is a separate experiment. Its metrics must not be merged into the TF-IDF learning curves.

• Before submission, ensure the report's stated class mapping, smoothing coefficient, data split policy, and exact run configuration are reproducible from the saved manifests.

## Conclusion

Human vote-distribution supervision lowers mean JS divergence in all 15 reported comparisons, but only DynaSent Round 2 at 300 labels survives Holm correction across the primary test family (adjusted p=0.0293). The appropriate conclusion is promising directional evidence with limited corrected significance—not universal superiority.